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Java SE 27 & JDK 27

Class FileTime

java.lang.Object
    java.nio.file.attribute.FileTime

All Implemented Interfaces:

Comparable<FileTime>

```
public final class FileTime
extends Object
implements Comparable<FileTime>
```

Represents the value of a file's time stamp attribute. For example, it may represent the time that the file was last `modified`, `accessed`, or `created`.

Instances of this class are immutable.

Since:

1.7

See Also:

`Files.setLastModifiedTime(Path, FileTime)`,
`Files.getLastModifiedTime(Path, LinkOption...)`

Method Summary

All Methods	Static Methods	

Modifier and Type	Method	Description
int	**compareTo**(FileTime other)	Compares the value of two FileTime objects for order.
boolean	**equals**(Object obj)	Tests this FileTime for equality with the given object.
static FileTime	**from**(long value, TimeUnit unit)	Returns a FileTime representing a value at the given unit of granularity.
static FileTime	**from**(Instant instant)	Returns a FileTime representing the same point of time value on the time-line as the provided Instant object.

static FileTime	**fromMillis**(long value)	Returns a FileTime representing the given value in milliseconds.
int	**hashCode**()	Computes a hash code for this file time.
long	**to**(TimeUnit unit)	Returns the value at the given unit of granularity.
Instant	**toInstant**()	Converts this FileTime object to an Instant.
long	**toMillis**()	Returns the value in milliseconds.
String	**toString**()	Returns the string representation of this FileTime.

Methods declared in class Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Method Details

from

```
public static FileTime from(long value,
                            TimeUnit unit)
```

Returns a FileTime representing a value at the given unit of granularity.

Parameters:

value - the value since the epoch (1970-01-01T00:00:00Z); can be negative

unit - the unit of granularity to interpret the value

Returns:

a FileTime representing the given value

fromMillis

```
public static FileTime fromMillis(long value)
```

Returns a FileTime representing the given value in milliseconds.

Parameters:

`value` - the value, in milliseconds, since the epoch (1970-01-01T00:00:00Z); can be negative

Returns:

a `FileTime` representing the given value

from

```
public static FileTime from(Instant instant)
```

Returns a `FileTime` representing the same point of time value on the time-line as the provided `Instant` object.

Parameters:

`instant` - the instant to convert

Returns:

a `FileTime` representing the same point on the time-line as the provided instant

Since:

1.8

to

```
public long to(TimeUnit unit)
```

Returns the value at the given unit of granularity.

Conversion from a coarser granularity that would numerically overflow saturate to `Long.MIN_VALUE` if negative or `Long.MAX_VALUE` if positive.

Parameters:

`unit` - the unit of granularity for the return value

Returns:

value in the given unit of granularity, since the epoch since the epoch (1970-01-01T00:00:00Z); can be negative

toMillis

```
public long toMillis()
```

Returns the value in milliseconds.

Conversion from a coarser granularity that would numerically overflow saturate to `Long.MIN_VALUE` if negative or `Long.MAX_VALUE` if positive.

Returns:

the value in milliseconds, since the epoch (1970-01-01T00:00:00Z)

toInstant

```
public Instant toInstant()
```

Converts this `FileTime` object to an `Instant`.

The conversion creates an `Instant` that represents the same point on the time-line as this `FileTime`.

`FileTime` can store points on the time-line further in the future and further in the past than `Instant`. Conversion from such further time points saturates to `Instant.MIN` if earlier than `Instant.MIN` or `Instant.MAX` if later than `Instant.MAX`.

Returns:

an instant representing the same point on the time-line as this `FileTime` object

Since:

1.8

equals

```
public boolean equals(Object obj)
```

Tests this `FileTime` for equality with the given object.

The result is `true` if and only if the argument is not `null` and is a `FileTime` that represents the same time. This method satisfies the general contract of the `Object.equals` method.

Overrides:

`equals` in class `Object`

Parameters:

`obj` - the object to compare with

Returns:

`true` if, and only if, the given object is a `FileTime` that represents the same time

See Also:

`Object.hashCode()`, `HashMap`

hashCode

```
public int hashCode()
```

Computes a hash code for this file time.

The hash code is based upon the value represented, and satisfies the general contract of the `Object.hashCode()` method.

Overrides:

`hashCode` in class `Object`

Returns:

the hash-code value

See Also:

`Object.equals(java.lang.Object)`, `System.identityHashCode(Object)`

compareTo

```
public int compareTo(FileTime other)
```

Compares the value of two `FileTime` objects for order.

Specified by:

`compareTo` in interface `Comparable<FileTime>`

Parameters:

`other` - the other `FileTime` to be compared

Returns:

`0` if this `FileTime` is equal to `other`, a value less than 0 if this `FileTime` represents a time that is before `other`, and a value greater than 0 if this `FileTime` represents a time that is after `other`

toString

```
public String toString()
```

Returns the string representation of this `FileTime`. The string is returned in the ISO 8601 format:

```
    YYYY-MM-DDThh:mm:ss[.s+]Z
```

where `"[.s+]"` represents a dot followed by one of more digits for the decimal fraction of a second. It is only present when the decimal fraction of a second is not zero. For example, `FileTime.fromMillis(1234567890000L).toString()` yields `"2009-02-13T23:31:30Z"`, and `FileTime.fromMillis(1234567890123L).toString()` yields `"2009-02-13T23:31:30.123Z"`.

A `FileTime` is primarily intended to represent the value of a file's time stamp. Where used to represent *extreme values*, where the year is less than `"0001"` or greater than `"9999"` then this method deviates from ISO 8601 in the same

manner as the XML Schema language. That is, the year may be expanded to more than four digits and may be negative-signed. If more than four digits then leading zeros are not present. The year before "0001" is "-0001".

Overrides:

`toString` in class `Object`

Returns:

the string representation of this file time

External Specifications

Date and Time Formats,
XML Schema Part 2: Datatypes Second Edition

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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